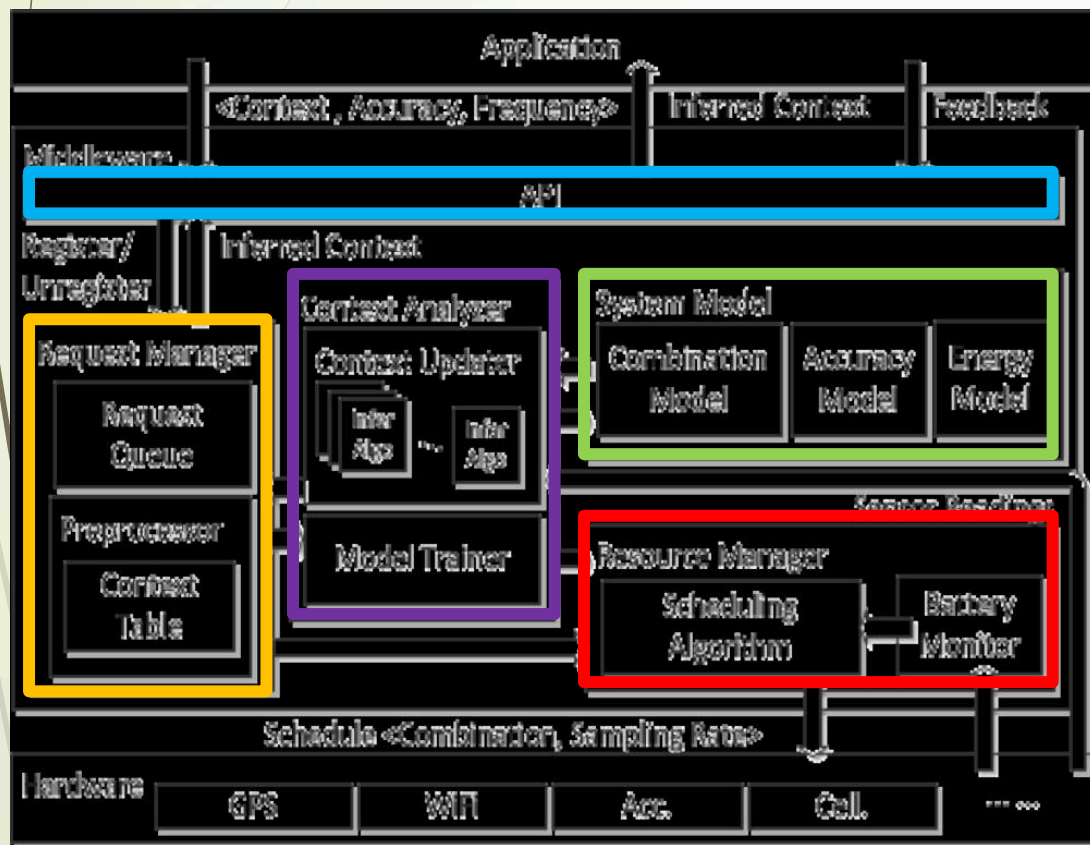


System Architecture



API:

1. Register()/Unregister()
2. Feedback()

Request Manager

1. Manages a Request Queue
2. Preprocess the requests

Context Analyzer

1. Context Updater
2. Model Trainer

Sensory Data

Infer Algo

Context

System Model

Combination/Accuracy/Energy

Resource Manager

1. Battery Monitor
2. **Scheduling Algorithm**

System Models

➤ **Combination model**

- The sensor combination of each inference algorithm

➤ **Accuracy model**

- The precision (accuracy) of each inference algorithm

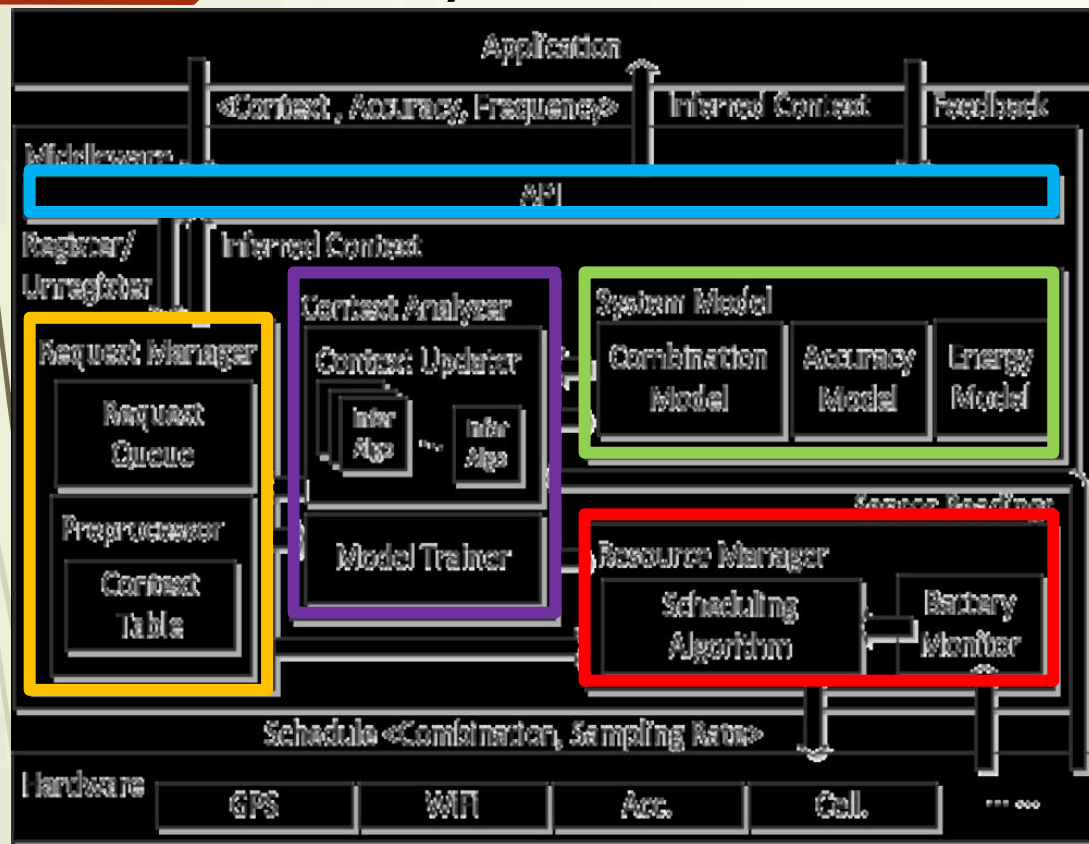
➤ **Energy model**

- The energy consumption of each sensor

➤ **Example:**

- InMeeting { <Acc. Mic. Wifi.> < 80% > <265mW> }

System Architecture



API:

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Sensory Data
Infer Algo
Context

System Model

Combination/Accuracy/Energy

Resource Manager

1. Battery Monitor
2. **Scheduling Algorithm**

**Coordinated and efficient sensor usage !*

**Avoid redundant energy waste !*

Scheduling Problem: Two Criteria

➤ **Energy Minimization (EM):**

Minimize the energy consumption

Satisfy all the apps' requirements

➤ **Accuracy Maximization (AM):**

Maximize the overall accuracy
within an energy budget



Problem Formulation

- Decision variable: $x_s \in \{0, 1\}$
 x_s indicates whether the sensor s should be activated
- Energy Minimization :

$$\min \sum_{s=1}^S e_s x_s$$

Minimize energy

$$\text{s.t. } \max_{c=1}^C \left\{ \left\lfloor \frac{\sum_{s=1}^S m_{c,s} x_s}{\sum_{s=1}^S m_{c,s}} \right\rfloor a_{c,r} \right\} \geq y_r, \forall r = 1, 2, \dots, R;$$

$$x_s \in \{0, 1\}, \forall s = 1, 2, \dots, S.$$

Satisfy all requirements



Problem Formulation

- Decision variable: $x_s \in \{0, 1\}$
 x_s indicates whether the sensor s should be activated
- Accuracy Maximization :

$$\max \sum_{r=1}^R \max_{c=1}^C \left\{ \left\lfloor \frac{\sum_{s=1}^S m_{c,s} x_s}{\sum_{s=1}^S m_{c,s}} \right\rfloor a_{c,r} \right\}$$

Maximize accuracy

$$s.t. \sum_{s=1}^S e_s x_s \leq E;$$

Energy budget

$$x_s \in \{0, 1\}, \forall s = 1, 2, \dots, S.$$



Proposed Scheduling Algorithms

- Optimal algorithms :
 - The formulations are “Integer Programming Problem”
 - We use a commercial optimization solver
 - Energy Minimization Algorithm (EMA)
Accuracy Maximization Algorithm (AMA)
 - + Good performance
 - - Only for small problems
- Heuristic algorithms :
 - Efficient Energy Minimization Algorithm (EEMA)
Efficient Accuracy Maximization Algorithm (EAMA)
 - + Shorter running time
 - - Suitable for smartphones

Efficient Energy Minimization Algorithm (EEMA)

- EEMA is inspired by the [Set Cover Problem](#)
- Define an utility function, and turn on the sensors that satisfy more requests with less energy cost

```

1: // Input:  $A, M, \hat{R}$ ; Output:  $X$ .
2: let  $X := \emptyset$ 
3: while  $\hat{R} \neq \emptyset, \max_{1 \leq c \leq C} g_c(A, M, X, \hat{R}) > 0$  do
4:   for each  $c = 1, 2, \dots, C$  do
5:     compute  $g_c(A, M, X, \hat{R})$  using Eq. (13)
6:   select  $c^* := \operatorname{argmax}_{c=1,2,\dots,C} g_c(A, M, X, \hat{R})$ 
7:    $X \leftarrow X \cup \{s | m_{c^*,s} = 1\}$ 
8:    $\hat{R} \leftarrow \hat{R} - \{r | y_r \leq a_{c^*,r}\}$ 

```

Calculate and Update the g_c

Choose the most efficiency combination

Efficient Accuracy Maximization Algorithm (EAMA)

- EAMA is inspired by the 0/1 knapsack Problem
- Define an utility function, and turn on the sensors that achieve higher accuracy with less energy cost

Calculate and Update the g'_c

Choose the most efficiency combination

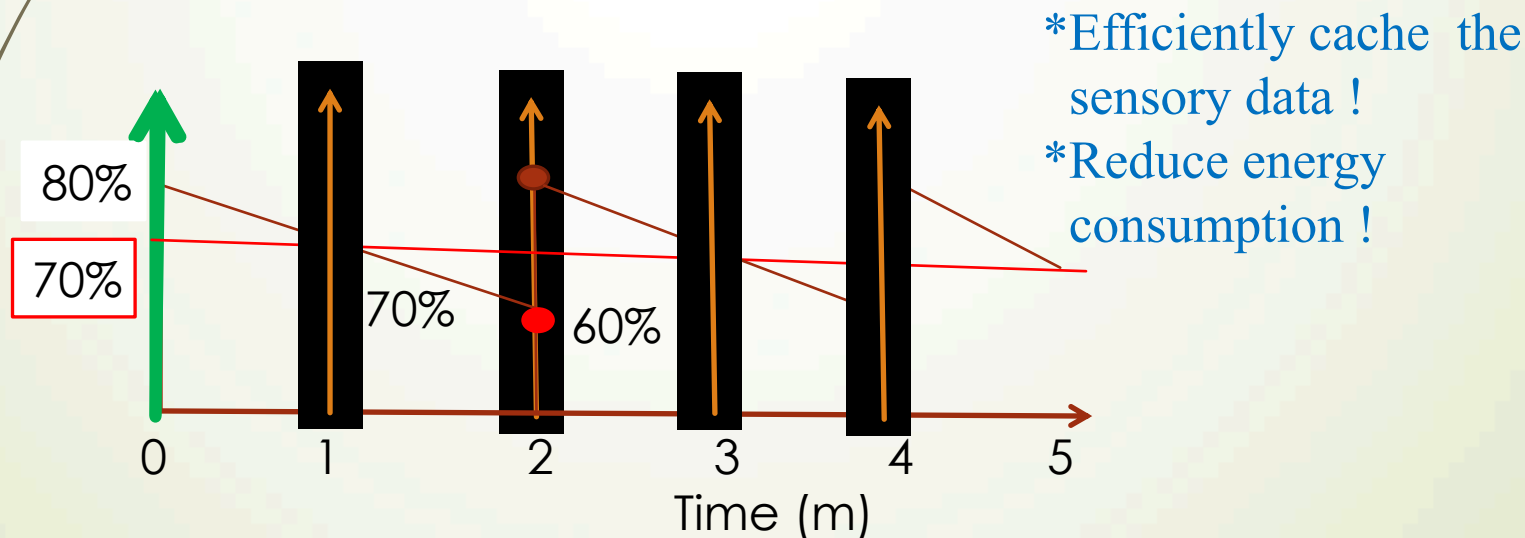
Check the energy budget

```

1: // Input:  $A, M$ ; Output:  $X$ 
2: let  $\alpha_X := 0, W := \{1, 2, \dots, O\}, \bar{Y} := 0$ 
3: while  $W \neq \emptyset$  do
4:   for each  $c := 1, 2, \dots, O$  do
5:     compute  $g'_c(A, M, X, \bar{Y})$  with Eq. (15)
6:   select  $c^* := \operatorname{argmax}_{c \in W} g'_c(A, M, X, \bar{Y})$ 
7:    $W \leftarrow W - \{c^*\}$ 
8:   if  $\alpha_X + w_{c^*}(M, X) \leq B$  then
9:      $X \leftarrow X \cup \{s | m_{c^*,s} = 1\}$ 
10:     $\alpha_X := \sum_{s=1}^S \alpha_{s,c^*}$ 
11:    for each  $r := 1, 2, \dots, R$  do
12:      if  $\bar{y}_r \leq \alpha_{c^*,r}$  then
13:         $\bar{y}_r := \alpha_{c^*,r}$ 
  
```

Heterogeneous Frequencies and Sampling Rates

- Extend the EEMA and the EAMA with heterogeneous frequencies/sampling rates as EEMA* and EAMA*
- Ex: a context requires 70 % accuracy, and the accuracy which decreases 10% per minute



Simulations

- Developed an event-driven simulator in Java
- Baseline algorithm :
 - Selects sensors with the highest accuracy for each context
- Compare scheduling algorithms :
 - Optimal : EMA / AMA
 - Efficient : EEMA / EAMA
 - With frequencies/sampling rates : EEMA* / EAMA*
 - Baseline

Data Collection

- Collect active apps in the Android activity stack from 5 users in three weeks
- Collect inference algorithms (from literature)
- Create the energy model



Energy	(mW)
Acc.	187
Blue.	195
WiFi	16
Mic.	62
GPS	546
Cell.	20

*Use Agilent 66321D battery emulator on a Galaxy Nexus phone

Parameters

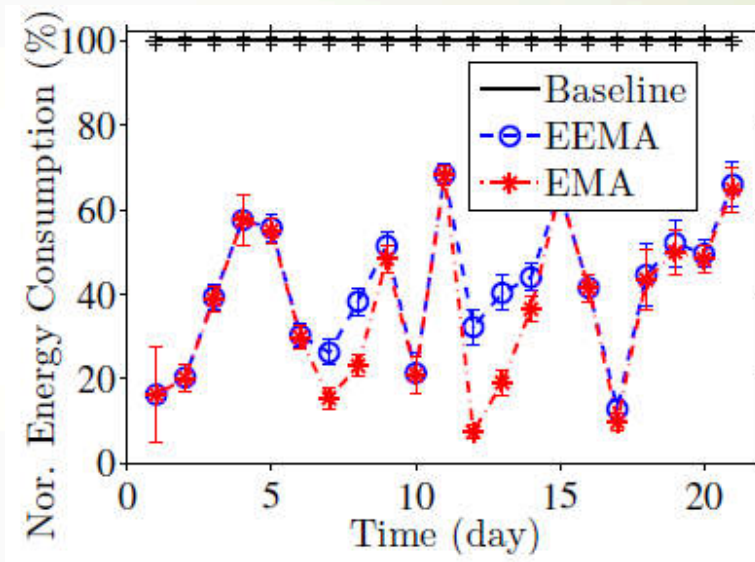
- Set a fixed scheduling time $T = 5\text{min}$
- Set E as the **Energy budget** in a scheduling time
 - $E = \{45, 52.5, 60, 67.5, 75\}$



Energy Saving in EM

➤ The average energy consumption with 5 users in 21 days

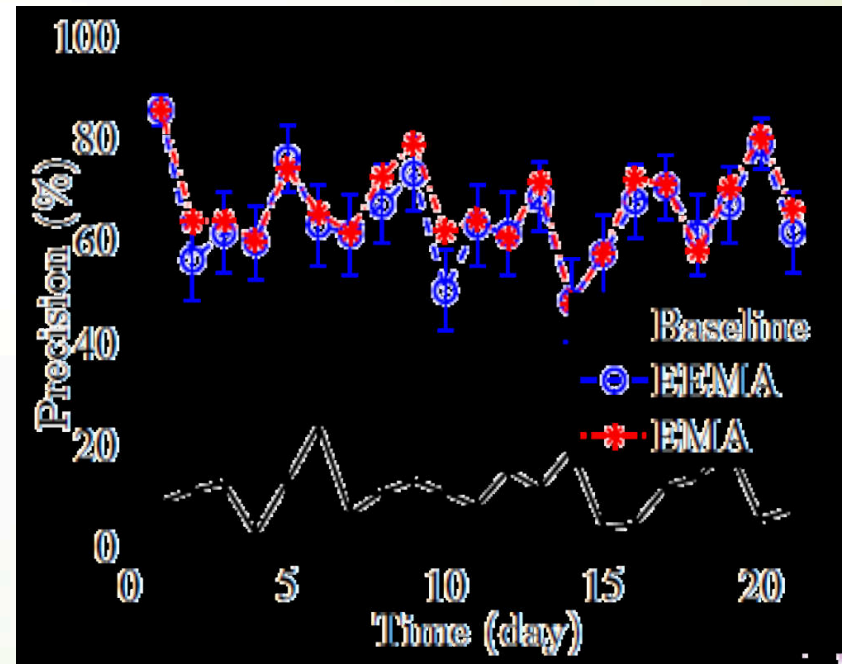
- * Saves at least 30%
- * EEMA achieves a small gap with EMA
- * EMA terminates in 50ms and EEMA terminates in 1ms



Improvement of Accuracy with Energy Minimization Criterion

➡ The accuracy with 5 users in 21 days

- * EEMA achieves at least **30%** higher accuracy
- * EEMA achieves a small gap with EMA



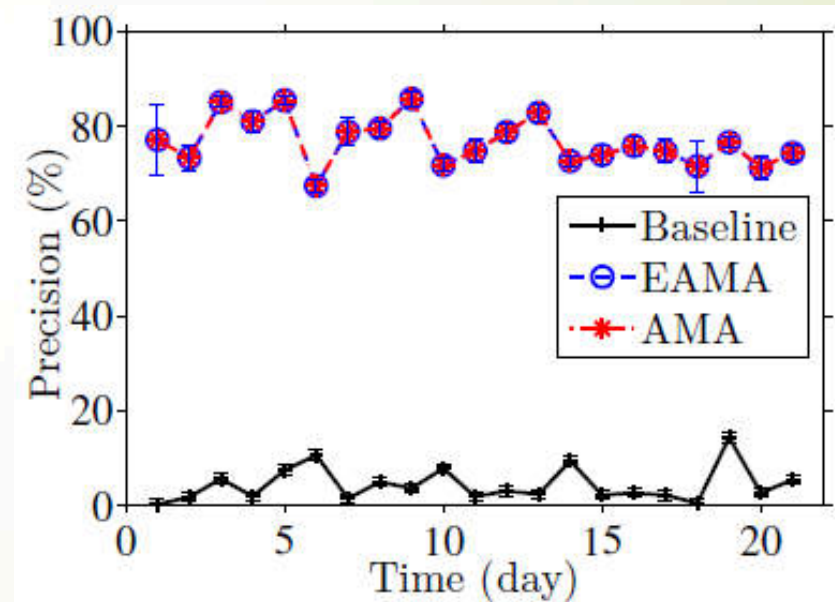
Accuracy Improvement with Accuracy Maximization Criterion

➤ The average accuracy with 5 users in 21 days

- * Accuracy is **72.38%** higher than the baseline

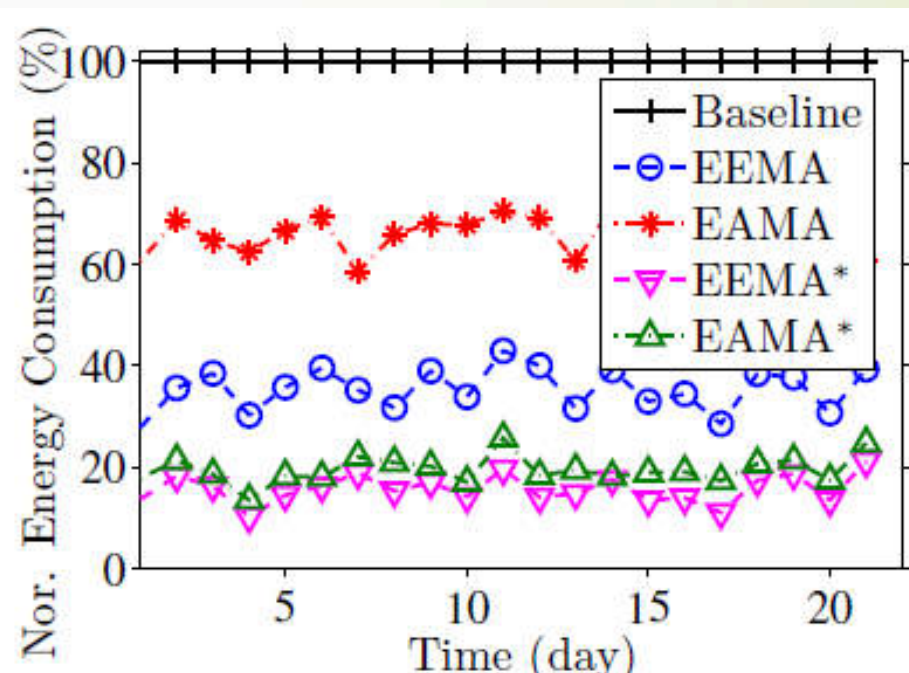
- * EAMA achieves a **~0.1%** gap with AMA

- * AMA terminates in **5 s** and EAMA terminates in **1ms**



Energy Saving with Heterogeneous Frequencies/Sampling Rates

- * EEMA saves 64.58%
EEMA* saves 84.66%
- * EAMA saves 33.68%
EAMA* saves 80.48%



- * Higher performance in EAMA*
=> high precision which is achieved by EAMA

Implementations on Android

- * EEMA :

Prolongs battery life **two** times

Achieves accuracy : **93.94%**

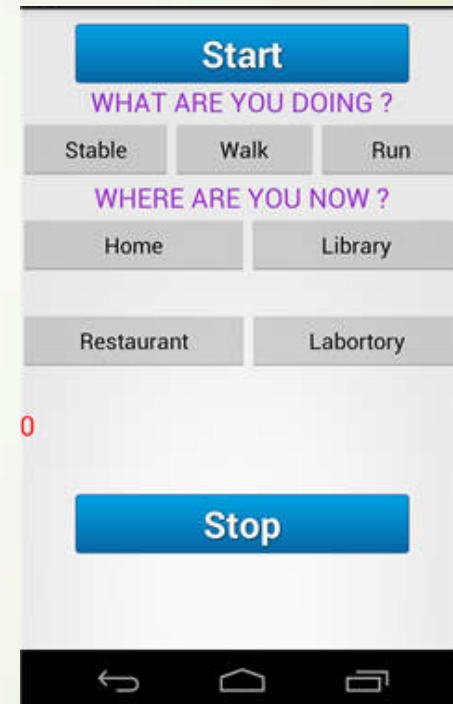
- * EAMA:

Prolongs battery life **1.5** times

Achieves accuracy : **94.85%**

- * Execution times :

EEMA and EAMA at most **306 ms** and **503 ms**



Questions

chsu@cs.nthu.edu.tw

